Sai Ravi Teja Gangavarapu

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Education

Carnegie Mellon University — Pittsburgh, PA
Master of Software Engineering - Scalable Systems (ML Systems, Statistics, QA, Formal Methods) — *Aug 2025 – Dec 2026*

University of Florida — Gainesville, FL
*Senior Cert + Master's coursework in Computer Science (Algorithms, UX, Graphics - **4.0**)* — *Jan 2024 – May 2024*

Skills

Python, C++, Go, TypeScript, Tanstack, SQL, Pydantic AI, CUDA, PyTorch, TF, React, Next.js, FastAPI, Langfuse, Docker, Kubernetes, AWS, PostgreSQL, Redis, ArgoCD, OpenAI, Cursor, Test and Spec-driven Development, Brownfield Refactoring, QA & Testing, Mathematics for ML, Deep Learning, Machine Learning, Statistics, Linear Algebra, MPI, HPC

Experience

Autonomy & Perception Engineer — CMU Lunabotics — Jan 2026 – Present

- Fused LiDAR + RGB-D for real-time mapping and trained segmentation models on synthetic data for obstacle classification. Built costmap and motion planning pipeline in ROS2 for autonomous rover navigation with sim-to-real validation.

Founding Software Engineer — Tapsta — Jul 2024 – Jun 2025

- Architected end-to-end full-stack social rewards mobile application using React, React Native, FastAPI, and PostgreSQL.
- Designed database architecture with 25+ schemas and optimized client-side operations with caching and debouncing, achieving 30% faster response times. Worked with product team to redesign onboarding flow, driving 40% user retention.
- Orchestrated Plaid API integration alongside REST API creation, establishing secure bank connectivity and enabling automated cashback processing, plus ACH transfers for over 1,500 application users.
- Made containerized CI/CD pipeline with Docker, AWS, and ArgoCD that reduced deployment time by 70%.
- Led development team of 3 SDE interns, establishing code review processes that contributed to **30%** faster feature delivery.
- Built a semantic people search engine for alumni networks using embedding-based similarity using LangGraph.

Research Assistant - Fan Lab, University of Florida — Feb 2024 – Jun 2024

- Applied genomic models for rare disease prediction using **transformer architectures**, processing **15,000+ DNA sequences**. Developed multi-layer perceptron models using protein embeddings (**ESM3**) analyzing RNA-binding proteins.

SDE Intern — Hashira.io — Apr 2023 – Dec 2023

- Taught LLMs to use APIs (incl. function calling), fine-tuning LLaMA/BERT in PyTorch to build a natural-language interface for payroll, achieving 90% natural language to API accuracy and reducing API integration time by 25%.
- Implemented full-stack crypto analytics platform using React, FastAPI, Postgres, Pandas, and Golang with real-time monitoring dashboard and a microservice for a leaderboard system for garden.finance, boosting user engagement by 30%.
- Improved backend performance using Go and PostgreSQL optimizations, supporting $150M+ trading volume over 30 days with 20% faster queries on garden.finance.

Research Assistant - Mahindra Ecole Centrale — Dec 2022 – Jan 2025

- Published research on emotion-targeted music generation using FFT, differential evolution, F C-means, self-organizing maps achieving 85% accuracy in classification (IEEE CEC 24). Implemented ALI-GAN model with t-SNE and PCA for clustering analysis on 1000+ samples, achieving unsupervised music genre classification. Built MIR pipelines extracting 100+ features.

Projects

Project RECON | Raspberry Pi, OpenMPI, GlusterFS | $2000 funding

- Architected 8-node Raspberry Pi 4B **compute cluster** with distributed storage, LDAP authentication, and Slurm job scheduling, serving 400+ students for coursework.

OneAIClick.com | Python, React.js, HuggingFace, PEFT/LoRA *(Built no-code/low code tool)*

- Developed an LLM fine-tuning abstraction tool for HuggingFace models that cut development cycles by 60% and reduced boilerplate code by 75%, enabling rapid idea validation and full data privacy.

CUDA Ray Tracer & Audio-Reactive Visualizer and **Graphics Engine** | C++, **CUDA**, OpenGL

- Built GPU-accelerated ray tracing system achieving **1,600x speedup** (7.5s to 0.0045s per frame) over CPU implementation, with real-time FFT-based audio reactive 256-agent boids simulation. Also, built modular 3D renderer in C++ and OpenGL

gitask.org | WebGPU, TypeScript, React | **200+ users**

- Offline, in-browser RAG tool to chat with any GitHub repo via URL swap (`github.com` → `gitask.org`). Local LLM inference via WebGPU. Did AST chunking, graph-aware hybrid search, binary quantized embeddings.

Leadership and Achievements

President, Enigma, the Computer Science Club (2021-2023): Conducted technical workshops on Gamedev, ML, Linux, heading outreach initiatives increasing the club membership by 40%, reaching over 2000 students. Collabs - Ubisoft, NVIDIA.
1st Place WaffleHacks 2024 (320+ participants): Built FlashFocus - a Chrome extension to block distractions and turn them into learning with AI flashcard quizzes using RAG system. (React, FastAPI, SQL, Groq)
1st Place NR Hackathon 2023 (50 teams): Engineered distributed cluster implementing Raft consensus protocol from scratch.
Winner AWAP@CMU 2026 (80 teams): Multi-agent kitchen solver. Hungarian algorithm, Cooperative A*.
Best Poster, CMU 10-799: Trained a flow matching model to generate realistic faces from rough sketches. Cut p-error by 50%.